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1. The line ℓ_1 is the intersection of the planes

$$x + y + z = 1$$

and

$$2x - y + z = 4$$

The plane Π has equation

$$x + y + 2z = 3$$

The line ℓ_2 is the reflection of the line ℓ_1 in the plane Π .

Find a vector equation of the line ℓ_2 .

[7]

2. The plane Π has vector equation

$$\mathbf{r} = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} + s \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} + t \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$$

The line ℓ_1 passes through the point $P(4, 1, 2)$ and is perpendicular to Π .

The line ℓ_1 meets Π at the point Q .

(a) Find the coordinates of Q . [4]

Given that the point $R(3, 1, 0)$ lies on Π ,

(b) find a Cartesian equation of the plane containing P , Q and R . [4]

The line ℓ_2 passes through P and R .

The line ℓ_3 is the reflection of ℓ_2 in Π .

(c) Find a vector equation of ℓ_3 . [4]

3. A plane passes through the points $C(1, -1, 2)$, $D(3, 0, 1)$ and $E(2, 2, 2)$.

The point A has coordinates $(4, -2, 7)$ and the point B is the reflection of A in the plane.

Find the coordinates of the point B .

[7]

4. A line ℓ has equation

$$\mathbf{r} = \begin{pmatrix} 4 \\ 2 \\ -3 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}$$

Another line m has equation

$$\mathbf{r} = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}$$

A plane Π contains m and is parallel to ℓ .

- (i) Find an equation of Π , giving your answer in the form $\mathbf{r} \cdot \mathbf{n} = p$. [4]
- (ii) Find the distance between ℓ and Π . [4]
- (iii) Find an equation of the line which is the reflection of ℓ in Π , giving your answer in the form $\mathbf{r} = \mathbf{a} + t\mathbf{b}$. [4]

5. The line ℓ_1 has vector equation

$$\mathbf{r} = \begin{pmatrix} 1 \\ 4 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}$$

The plane Π_1 passes through the point $(2, 0, 0)$ and has normal vector

$$\begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix}$$

(a) Find the point of intersection of ℓ_1 and Π_1 .

[2]

The line ℓ_2 is the reflection of the line ℓ_1 in the plane Π_1 .

(b) Show that a vector equation for the line ℓ_2 is

$$\mathbf{r} = \begin{pmatrix} 4 \\ 3 \\ 2 \end{pmatrix} + \mu \begin{pmatrix} -1 \\ 2 \\ -1 \end{pmatrix}$$

where μ is a scalar parameter.

[5]

The plane Π_2 contains the line ℓ_1 and the line ℓ_2 .

(c) Determine a vector equation for the line of intersection of Π_1 and Π_2 .

[2]

6. The triangle ABC lies in the plane Π . The vertices are

$$A(1, 0, 0), B(2, 2, 1) \text{ and } C(1, 1, c)$$

where c is a constant.

(a) Find, in terms of c , $\overrightarrow{BA} \times \overrightarrow{BC}$. [3]

Given that $\overrightarrow{BA} \times \overrightarrow{BC} = d\mathbf{i} + 2\mathbf{j} - \mathbf{k}$, where d is a constant,

(b) find the value of c and show that $d = -3$. [2]

(c) find an equation of Π in the form $\mathbf{r} \cdot \mathbf{n} = p$, where p is a constant. [3]

The point S has position vector $3\mathbf{i} + \mathbf{j} + 2\mathbf{k}$. The point S' is the image of S under reflection in Π .

(d) Find the position vector of S' . [5]

7. The line ℓ has equation

$$\mathbf{r} = (\mathbf{i} + \mathbf{k}) + \lambda(\mathbf{i} + 2\mathbf{j} + 3\mathbf{k})$$

where λ is a scalar parameter, and the plane Π has equation

$$\mathbf{r} \cdot (2\mathbf{i} - \mathbf{j} + \mathbf{k}) = 9$$

(a) Find the coordinates of the point of intersection of ℓ and Π . [4]

The perpendicular to Π from the point $A(0, 4, 1)$ meets Π at the point B .

(b) Verify that the coordinates of B are $(4, 2, 3)$. [3]

The point $A(0, 4, 1)$ is reflected in the plane Π to give the image point A' .

(c) Find the coordinates of the point A' . [2]

(d) Find an equation for the line obtained by reflecting the line ℓ in the plane Π , giving your answer in the form

$$\mathbf{r} \times \mathbf{a} = \mathbf{b} \quad [4]$$